EE 66
Fall 2026

Signals, Dynamics, and Information

Homework 8

This homework is due Friday, October 23, 2026 at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned). as well as your IPython notebook saved as a PDF.

If you do not attach a PDF "printout" of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

On occasion, a problem set contains one or more problems designated as optional. Your solutions to such problems are not to be evaluated, and will not count toward your grade. Nevertheless, you are still responsible for learning the subject matter within their scope. In other words, we may test you on their scope.

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1. (Optional) Properties of Eigenvalues and Determinants

Useful facts. The following properties may be helpful:

- If $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ for some nonzero vector $\mathbf{v}$, then $\mathbf{v}$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ.
- The characteristic polynomial of a square matrix $\mathbf{A}$ is $\det(\mathbf{A} - \lambda\mathbf{I})$.
- The eigenvalues of $\mathbf{A}$ are the roots of its characteristic polynomial.
- The determinant is defined only for square matrices.
- $\det(\mathbf{A}) = \det(\mathbf{A}^\top)$.
- $\det(\mathbf{A}\mathbf{B}) = \det(\mathbf{A})\det(\mathbf{B})$.
- $\det(\mathbf{A}) = 0$ if and only if $\mathbf{A}$ is not invertible.

Here are some short exercises, providing you the opportunity to practice applying these properties to prove new ones. Try to formulate a "proof" for each in your head before writing anything down! Doing so will solidify your linear algebra intuitions.

(a) Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n\times n}$. $\mathbf{A}$ is invertible, but $\mathbf{B}$ is not invertible. Show that neither $\mathbf{A}\mathbf{B}$ nor $\mathbf{B}\mathbf{A}$ is invertible.

(b) Consider a square matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ with eigenvalues $\lambda_1, \lambda_2, \cdots, \lambda_n$. Prove that the determinant of $\mathbf{A}$ equals the product of its eigenvalues. *Hint: write out the characteristic polynomial.*

(c) Explain why $\mathbf{A} \in \mathbb{R}^{n\times n}$ is not invertible if and only if $\mathbf{A}$ has an eigenvalue of 0.

(d) Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ and define $\mathbf{B} = c\mathbf{A}$ for $c \in \mathbb{R}$. Suppose λ is an eigenvalue of $\mathbf{A}$ with corresponding eigenvector $\mathbf{v}$. Show that $\mathbf{v}$ is also an eigenvector of $\mathbf{B}$ and find its corresponding eigenvalue μ.

(e) Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ and define $\mathbf{B} = \mathbf{A} + c\mathbf{I}$ for $c \in \mathbb{R}$. Suppose λ is an eigenvalue of $\mathbf{A}$ with corresponding eigenvector $\mathbf{v}$. Show that $\mathbf{v}$ is also an eigenvector of $\mathbf{B}$ and find its corresponding eigenvalue μ.

(f) Given that $\mathbf{A}$ is a square matrix, show that $\mathbf{A}$ and $\mathbf{A}^\top$ have the same eigenvalues.

(g) Show that the nonzero eigenvalues of $\mathbf{A}^\top\mathbf{A}$ are the same as the nonzero eigenvalues of $\mathbf{A}\mathbf{A}^\top$.

(h) Let $\mathbf{A}$ be an invertible matrix. Show that if λ is an eigenvalue of $\mathbf{A}$, then $\dfrac{1}{\lambda}$ is an eigenvalue of $\mathbf{A}^{-1}$.

2. (Optional) Reverse Mechanical Diagonalization

It may be helpful for this problem to recall that $\text{tr}(\mathbf{A}) = \sum \lambda_i$ and $\det(\mathbf{A}) = \prod \lambda_i$.

(a) Diagonalize the matrix $\mathbf{A}$ below. That is, find $\mathbf{V}$, $\mathbf{\Lambda}$, and $\mathbf{V}^{-1}$ such that $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$, where $\mathbf{\Lambda}$ is diagonal with the eigenvalues of $\mathbf{A}$ along its diagonal.

$$\mathbf{A} = \begin{bmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{3}{4} \end{bmatrix}$$

(b) Consider a 2×2 matrix $\mathbf{B}$ with the following eigenpairs:

$$\lambda_1 = 2 \text{ with eigenvector } \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \lambda_2 = -1 \text{ with eigenvector } \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Find $\mathbf{B}$. Then verify your answer by checking that $\mathbf{B}\mathbf{v}_1 = \lambda_1 \mathbf{v}_1$ and $\mathbf{B}\mathbf{v}_2 = \lambda_2 \mathbf{v}_2$.

(c) Consider the eigenpairs of $\mathbf{B}$. For each of the following potential modifications to these eigenpairs, determine whether the matrix $\mathbf{B}$ changes.

(i) You use $5\mathbf{v}_1$ in place of $\mathbf{v}_1$.

(ii) You swap both the columns of $\mathbf{V}$, and the diagonal entries of $\mathbf{\Lambda}$.

(iii) You are told instead that the second eigenvector is $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

3. Spectral Theorem for Real Symmetric Matrices

(a) Show that all eigenvalues λ of a real symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ are real (i.e., $\lambda = \lambda^* \in \mathbb{R}$).
Hint: Let $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ for a nonzero vector $\mathbf{v} \in \mathbb{C}^n$. Try evaluating the scalar quantity $\mathbf{v}^\top \mathbf{A}\mathbf{v}^*$ in two different ways by associating $\mathbf{A}$ with $\mathbf{v}$ on the left versus $\mathbf{v}^*$ on the right.

(b) Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a real symmetric matrix. Suppose $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^n$ are eigenvectors of $\mathbf{A}$ corresponding to two distinct real eigenvalues $\lambda_1 \neq \lambda_2$. Show that $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal ($\mathbf{v}_2^\top \mathbf{v}_1 = 0$), and explain why this implies that eigenvectors corresponding to distinct eigenvalues are linearly independent.
Hint: Consider two distinct eigenpairs $(\lambda_1, \mathbf{v}_1)$ and $(\lambda_2, \mathbf{v}_2)$. Try evaluating the expression $\mathbf{v}_2^\top \mathbf{A}\mathbf{v}_1$ in two different ways using matrix symmetry $\mathbf{A} = \mathbf{A}^\top$.

(c) Suppose $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric and has n distinct eigenvalues $\lambda_1, \ldots, \lambda_n$ with corresponding normalized eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ (such that $\|\mathbf{v}_i\| = 1$). Use your results from parts (a) and (b) to show that $\mathbf{A}$ can be written in the form $\mathbf{A} = \mathbf{V}\Lambda\mathbf{V}^\top$, where $\mathbf{V}$ is an orthogonal matrix and Λ is a diagonal matrix containing the eigenvalues.

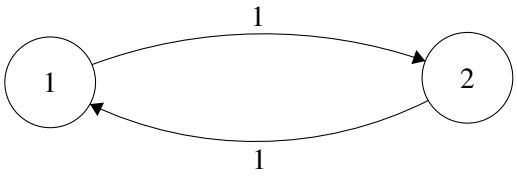
4. PageRank

In this problem, we consider using the adjacency matrices to represent "state transitions" in the setting of web traffic, and how users flow between different websites.

The “connectivity matrix”, $\mathbf{C}$, can be constructed using the state transition diagram as follows: entries c_{ji} represent the *proportion* of the people who are at website i that click the link for website j. This is slightly different from the adjacency matrix. Its values, c_{ji}, are the *proportion* of the people who are at website i that click the link for website j. It is associated with a graph that has labels stating the percentage (in our case of people) that is moving from one state to another.

Once computed, an eigenvector with eigenvalue 1 will have values which correspond to the steady-state frequency (i.e. the distribution in the limit as time goes to infinity) for the fraction of people for each webpage. When the elements of this eigenvector are made to **sum to one** (to conserve population), the ith element of the eigenvector will correspond to the fraction of people on the ith website.

(a) For graph A shown below, what are the steady-state frequencies (i.e. fraction of visitors in steady-state) for the two webpages? Graph A has weights in place to help you construct the connectivity matrix. Remember to ensure that your steady state-frequencies sum to 1 to maintain conservation.



Graph A

(b) In Graph A, notice that the system is periodic: a visitor at Page 1 at time n will always be at Page 2 at time $n+1$. This corresponds to the eigenvalue $\lambda = -1$ (an Nth root of unity where $N = 2$).

Using this observation, find the PageRank score vector, s^*.

Hint: Think about the mean of the oscillations over many steps.

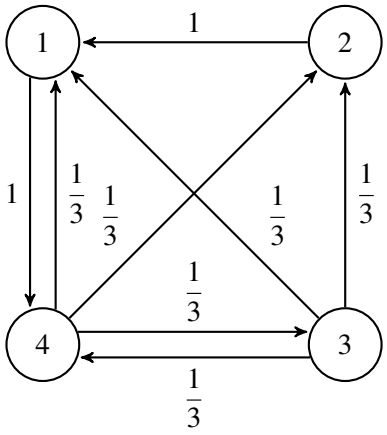
(c) For graph B, what are the steady-state frequencies for the webpages? You may use IPython and the Numpy command `numpy.linalg.eig` for this. Graph B is shown below, with weights in place to help you construct the connectivity matrix.

Hint: `numpy.linalg.eig` *returns eigenvectors and eigenvalues. The eigenvectors are arranged in a matrix in column-major order. In other words, given eigenvectors*

$\mathbf{v_1} = \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix}$ and $\mathbf{v_2} = \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix}$

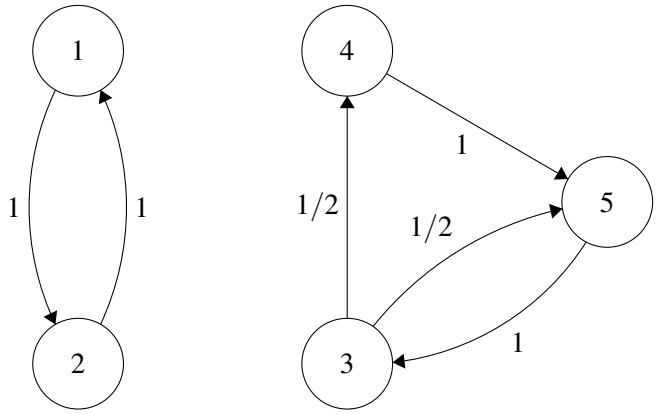
NumPy will return:

$$\begin{bmatrix} v_{11} & v_{21} \\ v_{12} & v_{22} \end{bmatrix} \tag{1}$$



Graph B

(d) Graph D with weights in place is shown below. Find the eigenspace that corresponds to the steady-state for graph D. How many independent systems (disjoint sets of webpages) are there in graph D versus in graph B? What is the dimension of the eigenspace corresponding to the steady-state for graph D? You may use IPython to compute the eigenvalues and eigenvectors again.



Graph D

5. Perpetual Motion Machine

For your science fair project, you are designing a **perpetual motion machine**, a machine designed to sustain its own motion for eternity, without external energy! You think you've come across the perfect design, but once you build it, you realize it isn't working as expected.

Your design involves the motion of a ball throughout a loosely defined path, which you believe should never slow down by **conservation of mechanical energy**. As a result, you thought, the ball should continue cycling through the path. However, after measuring its speed at a specific point on the path across multiple cycles, you realize that the ball is actually slowing down!

The ball's velocity, at the point of measurement, after the nth cycle is denoted by the vector $\mathbf{v}[n] = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$. In this question, a cycle refers to a single pass of the ball through the path, passing its initial starting point.

Note: Feedback control is beyond the scope of this course. *For this question, you only need to understand state space with inputs. If you're curious, you can learn more about feedback control in EE C128 or EECS C106A/B.*

(a) The initial velocity of the ball at the point of measurement is given by the vector $\mathbf{v}[0] = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. You take a few more measurements, and find the velocities after the first and second cycles are as follows:

$$\mathbf{v}[1] = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}, \mathbf{v}[2] = \begin{bmatrix} 0.25 \\ 0.6 \end{bmatrix}$$

Is the ball slowing down?

Hint: The magnitude of the velocity vector lets you know its speed.

(b) We can represent the relationship between the measurements from (a) as a difference equation of the form $\mathbf{v}[n+1] = \mathbf{A}\mathbf{v}[n]$. For the measurements shown above, **solve for A**.

(c) As you keep measuring the ball's velocity, you wonder whether it could ever settle at a non-zero velocity. Since you have a rough model of the system, you decide to analyze it mathematically.

Find the eigenvalues of A. Using this, demonstrate that, regardless of its initial velocity, the ball will always come to a complete stop over time.

Hint: Write $\mathbf{v}[n] = \mathbf{A}^n\mathbf{v}[0]$ *and use* $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$ *to examine the limit as* $n \to \infty$.

(d) Since the ball always slows to a stop on its own, keeping your "perpetual" motion machine alive requires external energy. You add a motor that boosts the ball's velocity at the point of measurement.

The motor power during the measurement after cycle n is denoted by the vector $\mathbf{u}[n]$, representing the effective power of the motor in the x and y directions.

The motor's effect on the velocity of the ball is a linear transformation of $\mathbf{u}[n]$, denoted by $\mathbf{B}\mathbf{u}[n]$, where $\mathbf{B}$ is a matrix. Now, our vector difference equation is as follows:

$$\mathbf{v}[n+1] = \mathbf{A}\mathbf{v}[n] + \mathbf{B}\mathbf{u}[n]$$

In an attempt to find a suitable input such that the speed of the ball remains constant, you initially try to use a constant motor power input $\mathbf{u}[n] = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$ for all n and $\mathbf{B} = \mathbf{I}$.

Compute the value of $\mathbf{v}[1]$, assuming $\mathbf{v}[0] = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Is this a suitable control input for your science fair project?

(e) Now, you get creative and try to adapt the motor input to the velocity of the ball. Assume we use the motor power input $\mathbf{u}[n] = \mathbf{v}[n]$ for all n and $\mathbf{B} = \mathbf{I}$. **Find $\|\mathbf{v}[n]\|$ as $n \to \infty$. Is this a suitable control input for your science fair project?**

(f) Now, you use a tuned **feedback control** input. Instead of a fixed boost, let the motor react to the ball: at each cycle it reads the ball's current velocity and sets its power from that reading, so the input is

$$\mathbf{u}[n] = \mathbf{F}\mathbf{v}[n] \text{ for all } n, \text{ where } \mathbf{F} = \begin{bmatrix} 0.5 & 0 \\ -0.5 & 0.3 \end{bmatrix}$$

Choosing the input from a measurement of the system is what makes this **feedback control**: the ball's velocity is fed back to the motor. These sorts of systems are a really interesting topic, and can be used in a wide variety of applications; in later classes, you'll learn to construct these inputs yourself!

Once again, the overall vector difference equation is still as follows:

$$\mathbf{v}[n+1] = \mathbf{A}\mathbf{v}[n] + \mathbf{B}\mathbf{u}[n]$$

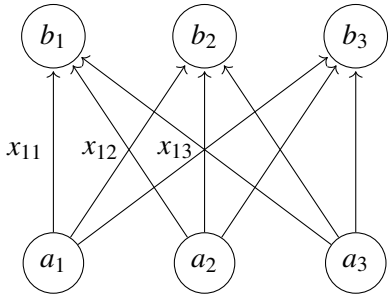
with $\mathbf{B} = \mathbf{I}$.

For this question, **show that, using the given control system, the measured velocity of the ball stays constant.**

6. Linear Neural Networks

Considered the pride and joy of modern-day machine learning, neural networks are simply the composition of basic mathematical functions that you already know about! In this problem, we will consider a very basic type of neural network, a *fully-connected linear neural network*, and derive closed-form solutions that describe the process used to train these networks to model complex behavior.

First, we will consider a single-layer neural network, in which a *linear combination* of the input values $a_1, a_2, \ldots, a_n$ are used to produce prediction values $b_1, b_2, \ldots, b_m$. An example of this architecture for $n = m = 3$ can be seen in the following figure:



In this case, we express mathematically that

$$b_j = \sum_{i=1}^{n} x_{ij} a_i$$

where each x_{ij} is the *weight* of the connection from input a_i to output b_j (note that only the weights from input a_1 are labeled in the diagram for visual reasons, but all input-output pairs do have appropriate weights). These model weights are the values that the model must "learn" from training examples in order to predict new outputs from future inputs.

(a) Using the provided mathematical expression, write b_j as the inner product between two vectors. Be sure to specify the entries of your vectors and what they represent.

(b) Let us define the vector $\mathbf{b} = \begin{bmatrix} b_1 & b_2 & \cdots & b_m \end{bmatrix}^\top$ (notice that this is a column vector). Using the inner product you derived above, express $\mathbf{b}$ as a matrix-vector multiplication. Be sure to specify the entries of your matrix and vector respectively, and describe what they represent. Note that this expression actually describes how we predict $\mathbf{b}$ given some inputs, a process known as *inference*.

If we actually want to use our neural network architecture, we first need to train it! In order to do this, we will estimate the weights x_{ij} in our network that best approximate the desired output $\mathbf{b}'$ given inputs $\{a'_1, a'_2, \ldots, a'_n\}$ (we can indicate these with the input vector $\mathbf{a}'$). It turns out that we'll need more than just one input-output pair to train our neural network, so let's define our dataset as the following pair of matrices:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mp} \end{bmatrix}$$

where $p \geq m, n$. For these matrices, a_{ki} is the value of a_i in the kth training input, and b_{jk} is the value of b_j in the kth training output. We can define each training pair $(\mathbf{a}_k, \mathbf{b}_k)$ as the kth **row** of matrix $\mathbf{A}$ and kth **column** of $\mathbf{B}$.

Assuming that our matrix-vector multiplication from the previous subpart can be written in the form $\mathbf{b} = \mathbf{X}^\top \mathbf{a}$, we want to find the matrix $\mathbf{X}$ that minimizes

$$\arg\min_{\mathbf{X} \in \mathbb{R}^{n \times m}} \|\mathbf{X}^\top \mathbf{A}^\top - \mathbf{B}\|_F^2$$

given our training data $(\mathbf{A}, \mathbf{B})$. The norm $\|\mathbf{M}\|_F$ of a matrix M is known as the *Frobenius norm* of that matrix.

As it turns out, this is actually the Least Squares problem, but with matrices instead! For this part, assume that the columns of $\mathbf{A}$ are linearly independent.

(c) Before we write an expression for the trained value of our entire matrix $\mathbf{X}$, let us consider a simpler version of the problem,

$$\arg\min_{\mathbf{x_j} \in \mathbb{R}^n} \|\mathbf{x}_j^\top \mathbf{A}^\top - \mathbf{b}_j^\top\|^2$$

where $\mathbf{x}_j$ is the jth column of $\mathbf{X}$ and $\mathbf{b}_j^\top$ is the jth row of our output matrix $\mathbf{B}$ (you can verify this by matching up dimensions). Use Least Squares to find an expression for this optimal vector of weights, $\mathbf{x}_j^*$, in terms of $\mathbf{A}$ and $\mathbf{b}_j$.

Afterwards, convince yourself how this shows that $\mathbf{x}_j^*$ is the weight vector that best maps the values of all the inputs to the output b_j of our network.

Hint: What is the relationship between the norm of a vector and the norm of its transpose?

(d) Now that we have derived a closed-form solution for a particular set of weights $\mathbf{x}_j^*$, derive a closed-form solution for the optimal value of the weight matrix, $\mathbf{X}^*$, in terms of $\mathbf{A}$ and $\mathbf{B}$.

It may help to know that $\|\mathbf{M}\|_F^2 = \sum_{i=1}^{n} \|\mathbf{m}_i\|_2^2$, where $\{\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_n\}$ are the rows of $\mathbf{M}$. In other words, the squared Frobenius norm of a matrix is equal to the sum of the squared ℓ_2 norms of its rows.

To learn more than you ever want to know about neural networks, take CS 189 and CS 182!

7. Homework Reflection

Doing homework is an **investment in your education**, and the course staff wants to hear about your experience with this problem set! After finishing the rest of the homework, fill out these brief reflections on what you got out of your time. We'll read this and take your answers into account as we write future assignments, in this semester and future ones.

Note that we are *genuinely curious* to read what you have to write. We expect *brief but thoughtful* responses. **Hence, this part will be weighed significantly in your homework completion grade.**

(a) Did you work on this homework alone or in a group? Did this work well for you? *We encourage you to attempt each problem on your own first, and to work with a group when you're stuck.*

(b) What resources did you use to solve the homework problems? How did you use these resources, and how did they help you? Feel free to mention LLM tools, if you've used any.

(c) What was the most fun / interesting problem on this homework? What was the least fun / un-interesting problem?

(d) Did you feel prepared to tackle these problems? Were there any problems that seemed completely disjoint from what you've seen in class? Was there anything covered in class that you weren't able practice in this assignment?

(e) Is there anything else you would like to share?